

Kartik Raikar

AI Engineer — Generative AI — LLM Applications

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI evaluation systems. Experienced in designing and developing full-stack AI applications using Python, FastAPI, React.js, SQL, REST APIs, and Git, with a strong focus on scalable, reliable, and production-ready AI solutions.

PROJECTS

AtlasOS – AI Memory Operating System | *Python, FastAPI, Next.js, PostgreSQL, Redis, Qdrant, Docker*

- Built a **multi-tenant AI Memory Operating System** using **FastAPI**, **PostgreSQL**, **Redis**, and **Qdrant**, enabling hierarchical memory management for AI agents and LLM applications.
- Implemented **PostgreSQL Row-Level Security (RLS)**, **vector search**, **semantic search**, and asynchronous pipelines with **PyTorch** and **Celery** for scalable AI infrastructure.

RagaAI Catalyst – Enterprise LLM Evaluation Platform | *Python, FastAPI, React.js, MongoDB*

- Developed a full-stack **LLM evaluation platform** to measure **faithfulness**, **relevance**, **toxicity**, and **correctness** across multiple AI models.
- Integrated **GPT-4**, **Claude**, **Gemini**, **Llama 3**, and **Mistral** through a unified **REST API** with automated benchmarking and real-time evaluation.

AI Debate Arena – Multi-Model Comparison Platform | *Python, FastAPI, React.js, JavaScript*

- Built a platform for **side-by-side LLM comparison** using a concurrent **FastAPI** backend for low-latency multi-model inference.
- Developed a **React.js** dashboard featuring live scoring, evaluation metrics, and automated verdict generation for model analysis.

NumPyGPT – GPT-Style Transformer Built from Scratch | *TypeScript*

- Implemented a **GPT-style Transformer** from scratch, including **multi-head attention**, **positional encoding**, **layer normalization**, and feed-forward networks.
- Built an **autoregressive text generation** pipeline with **top-k sampling**, temperature-controlled decoding, and attention visualization for transformer interpretability.

EDUCATION

B.E. Computer Science & Engineering (AI & ML)

Jain College of Engineering, Belagavi

Aug 2023 – Jul 2027

CGPA: 8.50/10

Pre-University Course (PCM)

Jain PU College, Belagavi

2021 – 2023

80%

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, React.js, Next.js, PyTorch

AI / ML: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Transformer Models, AI Evaluation, Prompt Engineering, Semantic Search, Vector Search

Databases: PostgreSQL, MongoDB, Redis, Qdrant

Developer Tools: Git, GitHub, Docker, Linux, Postman, VS Code

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- AWS Training & Certification – Fundamentals of Machine Learning and Artificial Intelligence
- Tata – GenAI Powered Data Analytics Job Simulation (Forage)

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Machine Learning